



Government Engineering College Palamu

Mathematics

B.Tech 2nd Semester

Assignment Problems-I

Subject: Engineering Mathematics-II (BSM02)

Jharkhand University of Technology (JUT)

Part A – MCQ Questions (1 Mark Each)

1. The order of the differential equation

$$y''' + (y')^2 + y = 0$$

is

- (A) 1
- (B) 2
- (C) 3
- (D) 4

2. The integrating factor of

$$\frac{dy}{dx} + Py = Q$$

is

- (A) e^P
- (B) $e^{\int P dx}$
- (C) $\int P dx$
- (D) Pe^x

3. A differential equation $Mdx + Ndy = 0$ is exact if

- (A) $M = N$

- (B) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$
(C) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
(D) None of these
4. The auxiliary equation of $y'' - 5y' + 6y = 0$ is
- (A) $m^2 + 5m + 6 = 0$
(B) $m^2 - 5m + 6 = 0$
(C) $m^2 - 6m + 5 = 0$
(D) $m^2 + 6m - 5 = 0$
5. Orthogonal trajectories intersect at
- (A) 0°
(B) 30°
(C) 45°
(D) 90°
6. The Wronskian is used to test:
- (A) Exactness
(B) Linear independence
(C) Orthogonality
(D) Homogeneity
7. Newton's law of cooling states that rate of cooling is proportional to
- (A) Absolute temperature
(B) Difference in temperature
(C) Product of temperatures
(D) Pressure
8. The differential equation of simple harmonic motion is
- (A) $y'' + y = 0$
(B) $y'' - y = 0$
(C) $y' + y = 0$
(D) $y'' = x$
9. The roots of the equation $m^2 - 5m + 6 = 0$ are
- (A) 1, 6
(B) 2, 3

- (C) -2, -3
(D) 5, 6
10. The complementary function is obtained from:
- (A) Homogeneous equation
(B) Non-homogeneous equation
(C) Exact equation
(D) Variable separable equation
11. The degree of the differential equation $(y'')^2 + y' = 0$ is
- (A) 1
(B) 2
(C) 3
(D) 4
12. Kirchhoff's voltage law is based on conservation of
- (A) Energy
(B) Charge
(C) Mass
(D) Momentum
13. The particular integral of $(D - 2)y = e^{2x}$ is
- (A) xe^{2x}
(B) e^{2x}
(C) $2e^{2x}$
(D) 0
14. A homogeneous differential equation can be solved using substitution
- (A) $y = vx$
(B) $y = v + x$
(C) $y = x + 1$
(D) $y = v^2$
15. The auxiliary equation corresponding to $y'' - 4y' + 4y = 0$ is
- (A) $m^2 + 4m + 4 = 0$
(B) $m^2 - 4m + 4 = 0$
(C) $m^2 - 4m - 4 = 0$
(D) $m^2 + 4m - 4 = 0$
16. Orthogonal trajectories intersect at angle

- (A) 0°
 - (B) 30°
 - (C) 45°
 - (D) 90°
17. The integrating factor of an exact differential equation is used to
- (A) Increase degree
 - (B) Make equation exact
 - (C) Reduce order
 - (D) Find roots
18. The solution of a homogeneous differential equation involves substitution
- (A) $x = vy$
 - (B) $y = vx$
 - (C) $y = v + x$
 - (D) $x = v + y$
19. The order of the equation $\left(\frac{d^3y}{dx^3}\right)^2 + \frac{dy}{dx} = 0$ is
- (A) 1
 - (B) 2
 - (C) 3
 - (D) 4
20. The degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3 + \frac{dy}{dx} = 0$ is
- (A) 1
 - (B) 2
 - (C) 3
 - (D) 4
21. Which method is suitable for solving $y'' + 5y' + 6y = e^x$?
- (A) Separation of variables
 - (B) Variation of parameters
 - (C) Linear differential equation method
 - (D) Constant coefficient method
22. The roots of $m^2 + 1 = 0$ are
- (A) $1, -1$
 - (B) $i, -i$

- (C) 0, 1
 - (D) 2, -2
23. The general solution of $y'' + y = 0$ contains
- (A) Exponential functions
 - (B) Polynomial functions
 - (C) Sine and cosine functions
 - (D) Logarithmic functions
24. In Newton's law of cooling, surrounding temperature is assumed to be
- (A) Variable
 - (B) Constant
 - (C) Infinite
 - (D) Zero
25. The equation $x^2y'' + xy' - y = 0$ is an example of
- (A) Linear equation
 - (B) Exact equation
 - (C) Cauchy–Euler equation
 - (D) Bernoulli equation
26. Which of the following represents simple harmonic motion
- (A) $y'' + ky = 0$
 - (B) $y'' - ky = 0$
 - (C) $y' + ky = 0$
 - (D) $y = kx$
27. The auxiliary equation helps to find:
- (A) Particular integral
 - (B) Complementary function
 - (C) Integrating factor
 - (D) Wronskian
28. A repeated root in auxiliary equation gives solution of form:
- (A) e^{mx}
 - (B) x^2e^{mx}
 - (C) $(c_1 + c_2x)e^{mx}$
 - (D) $c_1 + c_2$
29. The orthogonal trajectories of a family of curves are obtained by replacing

- (A) dy/dx by $-dx/dy$
 - (B) dy/dx by dx/dy
 - (C) dx/dy by dy/dx
 - (D) None of these
30. Variation of parameters is used to find:
- (A) Complementary function
 - (B) Particular integral
 - (C) Integrating factor
 - (D) Order of equation

Part B - Short Answer Questions

1. Solve

$$\frac{dy}{dx} = \frac{x+y}{x-y}$$

2. Solve the exact differential equation

$$(2xy + y^2)dx + (x^2 + 2xy)dy = 0$$

3. Solve the linear differential equation

$$\frac{dy}{dx} + 2y = e^{-x}$$

4. Find the orthogonal trajectories of

$$x^2 + y^2 = 2cx$$

5. Solve the Cauchy–Euler equation:

$$x^2 \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} + 4y = 0$$

6. Find the complementary function and particular integral of

$$(D^2 - 3D + 2)y = e^{2x}$$

7. State and apply Newton's law of cooling.
8. Form the differential equation governing current in an RL circuit.

Part C – Long Answer Questions

1. Solve the differential equation using variable separable method

$$\frac{dy}{dx} = \frac{x^2 + 1}{y^2 + 1}$$

2. Solve the homogeneous differential equation

$$\frac{dy}{dx} = \frac{x + y}{x}$$

3. Solve the exact differential equation:

$$(3x^2 + 2y)dx + (2x + 4y^3)dy = 0$$

4. Solve the higher-order differential equation

$$y'' + 5y' + 6y = e^{-2x}$$

5. Solve by variation of parameters

$$y'' + y = \tan x$$

6. Solve the Cauchy–Euler equation

$$x^2y'' + xy' - y = x^2$$

7. Obtain the orthogonal trajectories of

$$y = cx^2$$

8. A body cools from 90°C to 70°C in 10 minutes when surrounding temperature is 20°C . Find the temperature after 20 minutes.

9. A spring stretches 10 cm under a force of 20 N. A mass of 2 kg attached to the spring is displaced and released. Form the differential equation of motion and obtain the displacement equation.

10. Solve

$$\frac{dy}{dx} = \frac{x + y}{x - y}$$

11. Solve the exact differential equation

$$(2xy + y^2)dx + (x^2 + 2xy)dy = 0$$

12. Solve the Cauchy–Euler equation

$$x^2y'' - 3xy' + 4y = 0$$

13. Find the orthogonal trajectories of

$$x^2 + y^2 = 2cx$$

14. A body cools from 90°C to 70°C in 10 minutes when surrounding temperature is 20°C . Find the temperature after 20 minutes.

15. Solve the linear differential equation:

$$\frac{dy}{dx} + 2y = e^{-x}$$

16. Solve

$$y'' + 5y' + 6y = e^{-2x}$$

17. Solve by variation of parameters

$$y'' + y = \tan x$$

18. Obtain the orthogonal trajectories of

$$y = cx^2$$

19. Form the differential equation of simple harmonic motion for a spring–mass system.

20. Solve the homogeneous differential equation

$$\frac{dy}{dx} = \frac{x + y}{x}$$

21. Solve the exact differential equation

$$(3x^2 + 2y)dx + (2x + 4y^3)dy = 0$$

22. Find the complementary function and particular integral of

$$(D^2 - 3D + 2)y = e^{2x}$$

23. Solve the Cauchy–Euler equation

$$x^2y'' + xy' - y = x^2$$

24. State and apply Newton's law of cooling.